

HIGH-POWER DC-DC BOOST CONVERTER

Project Name:	High-Voltage DC Boost Converter	Controller IC:	UC3845AN (Current-Mode PWM)
Document Ref:	DOC-UC3845-BST-01	Topology:	Non-Isolated Peak Current Mode Boost

1. EXECUTIVE OVERVIEW & SPECIFICATIONS

This document provides comprehensive technical specifications, circuit analysis, component bill of materials (BOM), and operational instructions for the high-power DC-DC Boost Converter design based on the **UC3845AN** current-mode PWM control integrated circuit. This converter steps up low-voltage DC inputs (such as 12V battery banks or power supplies) to regulated high-voltage DC outputs up to 232V DC for high-voltage driving, capacitor charging, solar inverter buses, or industrial equipment testing.

PARAMETER	MIN VALUE	NOMINAL / TYPICAL	MAX VALUE	UNIT
Input Voltage (V_{IN})	9.0	12.0	15.0	V DC
Input Current Capacity	1.0	—	40.0	A
Output Voltage (V_{OUT})	9.0	Adjustable (via Pot)	232.0	V DC
Output Current (I_{OUT})	1.0	—	12.0	A
Output Voltage Capacitor Rating	200	220 μ F	400	V DC
Main Switch (MOSFET)	IRFP260N (200V, 50A, 0.040 Ω N-Channel Power MOSFET)			TO-247
Output Rectifier Diode	MUR3060PA (600V, 30A Ultrafast Power Rectifier)			TO-247
Inductor Value	30.0	47.0	60.0	μ H

HIGH VOLTAGE SAFETY WARNING

This boost converter circuit generates lethal output voltages up to **232V DC**. Output capacitors retain dangerous electrical charges even after power off. Always discharge the output bulk capacitor (220 μ F/400V) through a high-wattage bleed resistor before servicing. Wear insulated safety gear during operation.

2. CONTROLLER IC PIN CONFIGURATION (UC3845AN)

The **UC3845AN** is a fixed-frequency current-mode PWM controller designed specifically for off-line and DC-to-DC converter applications. It features an oscillator, high-gain error amplifier, current sensing comparator, and a high-current totem-pole output stage suitable for driving power MOSFETs directly.

PIN NO.	PIN NAME	CONNECTED COMPONENTS IN CIRCUIT	FUNCTIONAL DESCRIPTION
1	COMP	100k Ω Resistor + 102j (1nF) Capacitor filter network	Error amplifier output; used for feedback loop compensation network.
2	VFB	10k Potentiometer, 220 Ω & 33k Feedback Divider Network	Inverting input of Error Amplifier. Voltage feedback reference comparison (2.5V internally).
3	ISENSE	0.1 Ω Current Sense Resistor via 560 Ω filter	Current sense input. Compares switch current against error amplifier output for pulse limiting.
4	RT/CT	103j (10nF / 0.01 μ F) Timing Capacitor	Oscillator frequency set pin. Determines switching frequency in conjunction with timing resistor.
5	GND	Common Power & Signal Ground Rail	Circuit reference ground.
6	OUTPUT	10 Ω Gate Resistor to IRFP260N Gate	Totem-pole MOSFET gate driver output stage. Drives power switch ON/OFF.
7	VCC	4.7 Ω 2W Resistor + 1N4007 Diode from V_{IN}	Chip supply voltage input. Operating threshold UVLO enabled at 8.4V.
8	VREF	103j Decoupling Capacitor	Internal 5.0V Precision Reference Output.

3. BILL OF MATERIALS (BOM)

CATEGORY	COMPONENT REFERENCE	SPECIFICATION / VALUE	PACKAGE / FOOTPRINT	QTY
Integrated Circuit	IC1	UC3845AN Current Mode Controller	DIP-8	1
Power MOSFET	Q1	IRFP260N (200V / 50A / 0.04 Ω)	TO-247	1
Power Diode	D1	MUR3060PA Ultrafast Rectifier (600V / 30A)	TO-247 / TO-220	1
Signal Diode	D2	1N4007 General Purpose Rectifier (1000V / 1A)	DO-41	1
Inductor	L1	30 μ H – 60 μ H High-Current Core Inductor	Toroidal High-A Core	1
Capacitors	C1 (Input Filter)	2200 μ F Electrolytic Power Capacitor	Radial Lead	1
	C2 (Output Filter)	220 μ F / 200V–400V Electrolytic Capacitor	Radial Power Lead	1
	C3, C4, C5	103j (10nF / 0.01 μ F Film/Ceramic Capacitors)	Radial Ceramic/Film	3
	C6	102j (1nF / 0.001 μ F Ceramic Capacitor)	Radial Ceramic	1
Resistors	R1 (IC Power)		Axial Lead Power	1

CATEGORY	COMPONENT REFERENCE	SPECIFICATION / VALUE	PACKAGE / FOOTPRINT	QTY
		4.7 Ω / 2W Flameproof Power Resistor		
	R2 (Gate Driver)	10 Ω Carbon / Metal Film	1/4W Axial	1
	R3 (Current Sense)	0.1 Ω High-Power Current Sensing Resistor	2W-5W Ceramic/Shunt	1
	R4 (ISENSE Filter)	560 Ω Signal Resistor	1/4W Axial	1
	R5 (Feedback Upper)	33 k Ω High Voltage Feedback Resistor	1/2W Axial	1
	R6 (FB Divider)	220 Ω Resistor	1/4W Axial	1
	R7 (COMP Loop)	100 k Ω Resistor	1/4W Axial	1
	R8 (Ref Pullup)	10 k Ω Resistor	1/4W Axial	1
Potentiometer	VR1	10 k Ω Trimpot / Multi-turn Potentiometer	3-Pin Adjustable	1

4. DETAILED OPERATIONAL THEORY & CIRCUIT DESCRIPTION

4.1 Input Power Stage & Soft Power Delivery

The DC power source (9V–15V) enters the system across the main input terminals. A high-capacity **2200 μ F electrolytic capacitor** is connected directly across the DC input rails to suppress high-frequency ripple currents and smooth supply voltage dips caused by pulse switching. The power supply for controller IC (UC3845AN) Pin 7 (VCC) is conditioned through a **4.7 Ω 2W resistor** and a **1N4007 diode**. This prevents high-frequency switching noise on the power rail from feeding into the sensitive control logic of the IC.

4.2 Energy Storage & Boost Conversion Topology

The circuit operates on standard continuous/discontinuous peak current mode boost conversion:

- ON State:** When Pin 6 of UC3845AN drives the gate of MOSFET **IRFP260N** HIGH, the MOSFET turns ON. Current flows from the input terminal through the **30–60 μ H power inductor** to ground. Energy is stored in the magnetic field of the inductor ($E = \frac{1}{2} L I^2$). During this time, diode **MUR3060PA** is reverse biased, preventing output capacitor discharge into ground.
- OFF State:** When the IC switches the MOSFET OFF, the inductor's magnetic field collapses, inducing a reverse polarity voltage ($V_L = L \cdot di/dt$). This voltage adds in series with the input voltage ($V_{IN} + V_L$), forward-biasing the ultrafast rectifier diode **MUR3060PA** and transferring energy to the **220 μ F (200–400V)** output filter capacitor and connected load.

4.3 Current Sensing & Over-Current Protection

A **0.1 Ω current sense resistor** is connected between the Source terminal of the MOSFET and Ground. The voltage drop across this resistor is proportional to the switch current ($V_{SENSE} = I_{SW} \times 0.1\Omega$). This voltage is fed back to Pin 3 (ISENSE) of the UC3845AN through a **560 Ω RC filter**. When the voltage at Pin 3 reaches 1.0V, the controller terminates the current PWM cycle immediately, offering cycle-by-cycle over-current protection.

4.4 Output Voltage Feedback Loop

Output voltage regulation is achieved through a voltage divider connected between the high-voltage output rail and Pin 2 (VFB). The divider consists of a **33k Ω resistor**, a **220 Ω resistor**, and a **10k Ω potentiometer**. Adjusting the 10k potentiometer changes the division ratio, allowing the user to set the output voltage continuously anywhere between 9V and 232V DC.

Key Advantages of Current-Mode Control (UC3845AN)

Current-mode control provides inherent cycle-by-cycle pulse-width limit protection, superior line surge rejection, simplified control loop feedback compensation, and stable operation across wide load variations.

5. ASSEMBLY, THERMAL MANAGEMENT & TESTING STEPS

5.1 Heat Sinking Guidelines

- **IRFP260N MOSFET:** Dissipates considerable thermal energy during high-current operation. Mount Q1 on a heavy-duty aluminum heatsink with thermal paste and silicone insulator pads.
- **MUR3060PA Diode:** Subject to reverse recovery and forward voltage conduction losses under heavy high-voltage loads. Heat sinking is required when output currents exceed 2A continuously.

5.2 Commissioning & Initial Calibration

1. Verify all solder joints and check for shorts across high-voltage output rails.
2. Before initial powering, adjust the 10k potentiometer to its mid-point position.
3. Apply a low DC input voltage (e.g. 12V DC from a current-limited bench power supply set to 1A).
4. Measure the voltage across the 220 μ F output capacitor with a high-voltage rated digital multimeter.
5. Carefully rotate the 10k potentiometer to set the desired output voltage (9V to 232V DC).
6. Ensure output discharge resistors are present if connecting/disconnecting high voltage loads.